

IP Routing and Routing Protocols

Question 1: Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

Answer:

The router has three interfaces and each interface is the default gateway of one subnet. All subnets use the mask 255.255.255.0 (/24).

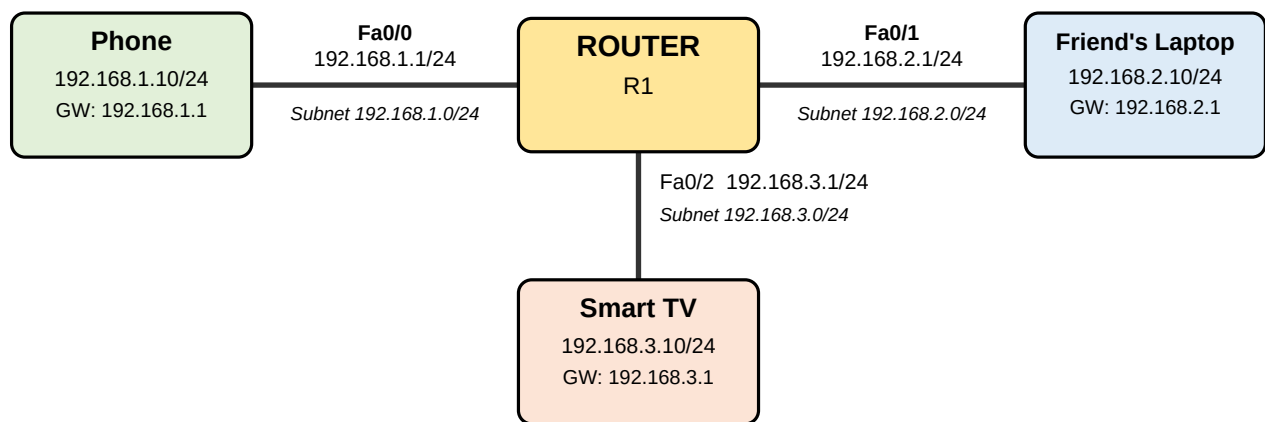


Figure 1: Phone, laptop and smart TV connected through a router

Question 2: Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Answer:

Assumed (made-up) network: My Laptop (192.168.1.10/24) - Router R1 - Router R2 - Friend's PC (192.168.2.20/24). R1 has Fa0/0 = 192.168.1.1/24 (my LAN) and Fa0/1 = 10.0.0.1/30 (link to R2). R2 has Fa0/1 = 10.0.0.2/30 and Fa0/0 = 192.168.2.1/24 (friend's LAN).

Step 1: Decision at my laptop. The laptop ANDs the destination IP 192.168.2.20 with its mask 255.255.255.0 and gets 192.168.2.0, which is different from its own network 192.168.1.0. The destination is therefore remote, so the packet is sent to the default gateway 192.168.1.1 (R1). The laptop uses ARP to find the MAC address of R1's Fa0/0 interface.

Laptop routing table	Network	Mask	Next hop	Interface
Connected	192.168.1.0	255.255.255.0	On-link	Ethernet
Default route	0.0.0.0	0.0.0.0	192.168.1.1	Ethernet

Step 2: Decision at R1. R1 receives the frame, removes the Layer 2 header, and looks at the destination IP 192.168.2.20. It checks its routing table from the most specific (longest prefix) match:

R1 routing table	Network / prefix	Next hop	Interface
C (connected)	192.168.1.0/24	Directly connected	Fa0/0
C (connected)	10.0.0.0/30	Directly connected	Fa0/1
S (static)	192.168.2.0/24	10.0.0.2	Fa0/1

192.168.2.20 matches only 192.168.2.0/24, so R1 decreases the TTL by 1, ARPs for 10.0.0.2 and forwards the packet out of Fa0/1 to R2.

Step 3: Decision at R2. R2 checks its table:

R2 routing table	Network / prefix	Next hop	Interface
C (connected)	10.0.0.0/30	Directly connected	Fa0/1
C (connected)	192.168.2.0/24	Directly connected	Fa0/0
S (static)	192.168.1.0/24	10.0.0.1	Fa0/1

The destination 192.168.2.20 matches the directly connected network 192.168.2.0/24, so R2 decreases the TTL again, uses ARP to find the MAC of the friend's PC and delivers the frame out of Fa0/0.

Step 4: Delivery. The friend's PC receives the packet, sees its own IP address as the destination and processes it. The reply follows the reverse path (PC to R2 to R1 to laptop) using the static route 192.168.1.0/24 on R2.

How the addresses change at each hop: the IP addresses stay the same from end to end, but the MAC addresses and TTL change at every hop (MAC addresses are shortened, made-up values).

Hop	Source IP	Destination IP	Source MAC	Dest. MAC	TTL
Laptop to R1	192.168.1.10	192.168.2.20	00:AA	00:B1 (R1 Fa0/0)	128
R1 to R2	192.168.1.10	192.168.2.20	00:B2 (R1 Fa0/1)	00:C2 (R2 Fa0/1)	127
R2 to PC	192.168.1.10	192.168.2.20	00:C1 (R2 Fa0/0)	00:DD (PC)	126

Question 3: Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Answer:

Fictional network: Mumbai is the head-office branch and connects to the Delhi and Bangalore branches through point-to-point WAN links. The Mumbai router is also connected to the Internet.

Branch	LAN subnet	WAN link	Router WAN address
Mumbai (head office)	10.1.1.0/24	Mumbai - Delhi: 10.0.0.0/30	Mumbai 10.0.0.1, Delhi 10.0.0.2
Delhi	10.1.2.0/24	Mumbai - Bangalore: 10.0.0.4/30	Mumbai 10.0.0.5, Bangalore 10.0.0.6
Bangalore	10.1.3.0/24		

Routing table of the Mumbai router:

Type	Network address	Subnet mask	Next hop	Interface
C	10.1.1.0	255.255.255.0	Directly connected	Gi0/0 (Mumbai LAN)
C	10.0.0.0	255.255.255.252	Directly connected	Se0/0/0 (link to Delhi)
C	10.0.0.4	255.255.255.252	Directly connected	Se0/0/1 (link to Bangalore)
S	10.1.2.0	255.255.255.0	10.0.0.2	Se0/0/0
S	10.1.3.0	255.255.255.0	10.0.0.6	Se0/0/1
S*	0.0.0.0	0.0.0.0	203.0.113.1 (ISP)	Gi0/1 (Internet)

C = connected route, S = static route, S* = static default route. Delhi and Bangalore routers use a default route (0.0.0.0/0) towards Mumbai through 10.0.0.1 and 10.0.0.5 respectively.

Question 4: Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Answer:

Static routing means the administrator manually enters every route. **Dynamic routing** means routers learn and update routes automatically using protocols such as RIP (hop-count based, maximum 15 hops) or OSPF (link-state, cost based).

Static routing

Pros	Cons
1. No CPU, memory or bandwidth is used for routing updates. Example: a home Wi-Fi router just uses one default route to the ISP.	1. Does not scale. Adding a new city branch to a delivery app's network means editing every router by hand.
2. Simple and predictable; easy to understand and troubleshoot in small networks (e.g. a small shop with one link).	2. No automatic failover. If a link goes down, traffic keeps going to the dead path and the app's orders may fail until an admin fixes it.
3. More secure because no routing information is advertised, so there is less chance of fake route updates.	3. Manual entry is error-prone; one wrong next hop can cause a routing loop or an outage.

Dynamic routing (RIP / OSPF)

Pros	Cons
1. Automatically adapts to link failures. Example: if the Mumbai-Delhi link fails, a food delivery app's traffic is rerouted through Bangalore without human action, similar to how a delivery rider app picks another road during a traffic jam.	1. Uses router CPU, memory and bandwidth to exchange routing updates.
2. Scales easily; new networks, such as a new city branch or a new college building, are learned automatically.	2. More complex to configure and troubleshoot (areas, costs, neighbours, authentication).

Pros	Cons
3. Chooses the best path (OSPF uses cost) and can balance load over equal paths, which helps busy services such as video streaming or online payments.	3. Convergence takes some time after a change, and unauthenticated updates can be attacked with false routes.

Question 5: Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Answer:

The following Cisco IOS configuration was generated using an AI tool for the two routers R1 and R2 from Question 2, with all networks placed in OSPF area 0. Each line is annotated.

Router R1

Configuration line	What it does
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode so that changes can be made.
hostname R1	Sets the name of the router to R1.
interface FastEthernet0/0	Selects the interface that faces the LAN for configuration.
ip address 192.168.1.1 255.255.255.0	Gives the LAN interface its IP address and subnet mask (gateway for the laptop).
no shutdown	Turns the interface on (interfaces are off by default on routers).
interface FastEthernet0/1	Selects the interface that connects to R2.
ip address 10.0.0.1 255.255.255.252	Assigns the IP address for the point-to-point link to R2.
no shutdown	Turns on the link interface.
router ospf 1	Starts the OSPF process with process ID 1 (the ID is only locally significant).
router-id 1.1.1.1	Sets a unique ID that identifies R1 to other OSPF routers.
network 192.168.1.0 0.0.0.255 area 0	Enables OSPF on the interface(s) in 192.168.1.0/24 and advertises this network in area 0. 0.0.0.255 is the wildcard mask (opposite of a subnet mask).
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the link to R2 (10.0.0.0/30) so that R1 can form a neighbour relationship with R2.
passive-interface FastEthernet0/0	Stops OSPF hello packets on the LAN, since no router exists there, but the network is still advertised.
end	Returns to privileged EXEC mode.
write memory	Saves the running configuration to the startup configuration.

Router R2

Configuration line	What it does
hostname R2	Sets the name of the router to R2.

Configuration line	What it does
interface FastEthernet0/1	Selects the interface that connects to R1.
ip address 10.0.0.2 255.255.255.252	Assigns R2's address on the link to R1.
no shutdown	Turns on the interface.
interface FastEthernet0/0	Selects the interface facing the friend's LAN.
ip address 192.168.2.1 255.255.255.0	Assigns the gateway address of the friend's LAN.
no shutdown	Turns on the interface.
router ospf 1	Starts the OSPF process on R2.
router-id 2.2.2.2	Sets a unique router ID for R2.
network 192.168.2.0 0.0.0.255 area 0	Advertises the friend's LAN in area 0.
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the link to R1 in area 0 (must be the same area as R1).
passive-interface FastEthernet0/0	Stops OSPF hellos on the LAN side.
end / write memory	Exits configuration mode and saves the configuration.

Verification commands:

Command	What it shows
show ip ospf neighbor	Whether R1 and R2 have become OSPF neighbours (state FULL).
show ip route ospf	Routes learned through OSPF, marked with the letter O.
ping 192.168.2.20	Tests end-to-end connectivity from R1's side to the friend's PC.